

Technique- Post Training Quantization (PTQ)

Model- MobileNetV2

Original model size- 14MB, 3.5 million parameters

Changes at model level- converts 32 bit floating point weights and activations to 8 bit integers(INT8)

Expected effect- reduces size by 4x, significantly increases inference speed with minimal accuracy drop

Why is it appropriate for edge?

1. MobileNetV2 is designed for mobile and embedded vision tasks where memory is constrained quantization allows the model to fit into much smaller RAM buffers.
2. Mobile hardware (like DSPs and NPUs) is heavily optimized for INT8 arithmetic, thus increasing hardware acceleration.
3. on-device inference without needing a constant network connection, preserving privacy and reducing latency.
4. PTQ (Post-Training Quantization) is highly efficient as it does not require retraining the model, making it ideal for rapid deployment cycles.
5. Accuracy while drastically reducing power consumption is critical for battery-powered edge devices.